

NEXTGEN PYTHON INTERNSHIP MX · Q2 2026 · STAGE 2

Workshop NoSQL DB MongoDB

Case study: how Forbes rebuilt their CMS on MongoDB

Follow-along session

Hands-on, all levels welcome

Stage 2 · Workshop 1 of 2

github.com/agent3dev/workshop-nosql-stage2-mongodb

FROM STAGE 1 TO STAGE 2

Same skills, a different mental model

STAGE 1 · POSTGRESQL

- Rows in fixed-column **tables**
- Schema changes need a **migration**
- Relationships resolved with **JOINS**
- One shape per table, always

STAGE 2 · MONGODB

- **Documents** in flexible collections
- New shapes need **zero migration**
- Relationships: **embed** or **reference**
- Different shapes, same collection

Why a magazine rewrote its entire CMS

- Forbes needed to publish **articles, videos, slideshows, and sponsored posts** — each shaped completely differently
- A relational schema meant either one giant table full of NULLs, or a new migration for every new content type
- They rebuilt the CMS on MongoDB and shipped it in **2 months**

“MongoDB’s document model meant developers could build new features quickly, easily incorporate changes, and better handle a growing diversity of data types.”

TODAY, WE REBUILD IT

Curriculum roadmap

#	FILE	TOPIC	THE FORBES NEED IT ANSWERS
01	01_crud.js	CRUD basics	Get comfortable reading/writing documents first
02	02_schema_design.js	Schema design	4 content types, 1 collection, no migrations — the actual reason Forbes chose Mongo
03	03_aggregation.js	Aggregation	“Real-time insight into social sharing” — Forbes’ own words
04	04_indexes.js	Indexes	Serving ~121M monthly users without a collection scan

CRUD basics

- Read: `find()` , `findOne()` , filters, sort, limit, count
- Write: `insertOne` / `insertMany` , then `updateOne` with `$set` , `$push` , `$inc`
- No JOIN: resolve `author_id` yourself, query inside an embedded array
- Goal isn't the syntax — it's getting comfortable before Exercise 2's real payoff

Schema design: embed vs. reference

- Same collection, four shapes — article, video, slideshow, sponsored — **zero migrations**
- **Embed** comments (bounded, always read with the article) vs. **reference** authors (one author, thousands of articles)
- `slug` is the one field every shape keeps fixed — Forbes' own "stable API on fluid data" idea, in miniature
- `$jsonSchema` validation: MongoDB is schema-**flexible**, not schema-less

Aggregation pipelines

- `find()` can't compute "average shares per tag" — that needs `$group`
- Top articles by shares, engagement by content type, author stats via `$lookup`
- Comment-level analysis with `$unwind` and `$size`
- This is Forbes' own words: "**real-time insight into social sharing**"

Indexes

- `explain()` first — see **COLLSCAN** become **IXSCAN**
- Unique index on `slug` — two articles can never collide on a URL
- Compound index `(type, published_at)` — matches the real "latest by type" query
- Text index for on-site search, with a relevance score

FORMAT

How today works

- This is a **follow-along-with-teacher** session, not a self-paced tutorial
- Some of you will type every command with me — others will watch and jump in on the open challenges. **Both are fine.**
- Each file ends with take-home challenges — we may not finish everything live, and that's expected
- Ask questions any time — we'll pause

BEFORE WE START

Get the stack running

- `git clone github.com/agent3dev/workshop-nosql-stage2-mongodb`
- `docker compose up -d`
- Visual UI: **mongo-express** at `localhost:8081`
- Or terminal: `docker exec -it forbes_mongo mongosh ...`

Full setup steps are in the repo README

LET'S BUILD

Rebuilding **Forbes' CMS**, one exercise at a time

github.com/agent3dev/workshop-nosql-stage2-mongodb